UV Bench, Rev C: design report

A 365 nm LED held at constant optical power by a photodiode and a microcontroller · project 01 of the electronics workspace · October 2026 · status: boards and parts ordered

1. Goal

A small bench light source for UV-fluorescence experiments. A ~1 W, 365 nm LED must deliver a **stable, settable light power**, not just a stable current: LED output drifts with temperature and age. A photodiode watches the LED and the microcontroller adjusts the drive until the reading matches the setpoint. An external board sets the power over I2C; a knob allows manual use. The board must be cheap, small and **solderable by hand**.

58 × 32 mm	**26 parts**	**≤ 0.33 A**	**5 V, I2C**
2 layers, Ø 22 mm cut-out for the LED	16 types, 4 resistor values, 2 capacitor values	LED current, limited in hardware	one JST-PH cable to the host board

2. How it works

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5 V ──► LED+ ──[ARM switch]──► UV LED ──► Q1 (MOSFET) ──► R1 3.9 Ω ──► GND
                                          ▲ gate              └──► I_MON (current, monitoring)
MCU PWM 20 kHz ──► RC filter (1 kΩ / 10 µF) ──┘
MCU ADC ◄── photodiode D1 + 100 kΩ load ◄~~~ a fraction of the LED light   (the feedback loop)
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- The MCU outputs PWM; an RC filter smooths it into a DC gate voltage (0–3.3 V, 4 mV ripple).
- Q1 with R1 forms a **voltage-controlled current sink**: $I \approx (V_{gate} - V_{GS}) / R1$. The LED current is DC, so the light has no flicker.
- The photodiode current (≈ 0.1 A/W at 365 nm) becomes a voltage on a resistor; the MCU compares it with the setpoint and corrects the PWM, in firmware.

3. Main design choices

- **One light loop, no current loop.** Rev A had an op-amp holding the current exactly *and* the MCU holding the light: two loops in cascade, 71 parts, 9 ICs. The MOSFET's current is imprecise and drifts, but the photodiode loop corrects any drift whatever its cause (MOSFET, LED ageing, LED temperature). So the op-amps, switch, AND gate and compensation network all went: **71 → 21 parts** (Rev B).
- **A hardware current limit for free.** At full drive Q1 is fully on and only R1 and the LED limit the current: $I_{max} = (5\ V - V_f) / 3.9\ \Omega \approx 0.33$ A. A firmware bug can turn the LED fully on, never overdrive it.
- **A resistor instead of a transimpedance amplifier.** The light is DC, so a plain 100 kΩ load is enough. The reverse bias keeps the photodiode linear. The firmware subtracts a dark reading taken at each switch-on.
- **Local regulation.** The loop runs on the board: the host only says "hold this power". If the I2C cable or the host hangs, the bench keeps regulating and can switch itself off.
- **Safety independent of the MCU.** A latching **UV ARM toggle switch** in the LED+ wire disconnects the LED physically, even if Q1 fails short. The firmware sees $I_{MON} \approx 0$ and reports ARM_OFF.
- **Optics from a torch.** The LED star (20 mm), reflector and ZWB2 filter come from a Convoy S2 365 nm flashlight. The star sits in the board's Ø 22 mm cut-out, screwed to an aluminium plate that serves as heatsink.
- **Rev C: a remote sensor on the target.** A second photodiode on a 12 × 21 mm head, under the same ZWB2 filter, on a 50 cm cable. It measures what the sample actually receives (≈ 5.6 µA per mW/cm²; 5–300 µA expected), so no amplifier is needed: a 10 kΩ load gives 0.06–2.8 V. The firmware can regulate on it or just report it.
- **Designed for hand soldering and learning.** 0805 passives, SOT-23 and TSSOP-20 packages, few distinct values, values printed on the assembly drawing. The schematic and board are generated by Python scripts; the routing is done with Freerouting, then checked by KiCad's ERC and DRC (0 errors).

4. Status and next steps

- **Done:** ERC 0 / DRC 0 on both boards; Gerbers ordered at JLCPCB (5 + 5, lead-free HASL, tag `uv-bench-revC-ordered`); parts ordered at DigiKey (~€53); Convoy S2 ordered; I2C interface spec v2.3.
- **Next:** check the star fits the cut-out and the MCP1702 pinout on arrival; firmware (CubeMX setup, the loop, I2C registers); a simulation of the LED driver; assembly and first power-up, starting current-limited.

5. The circuit

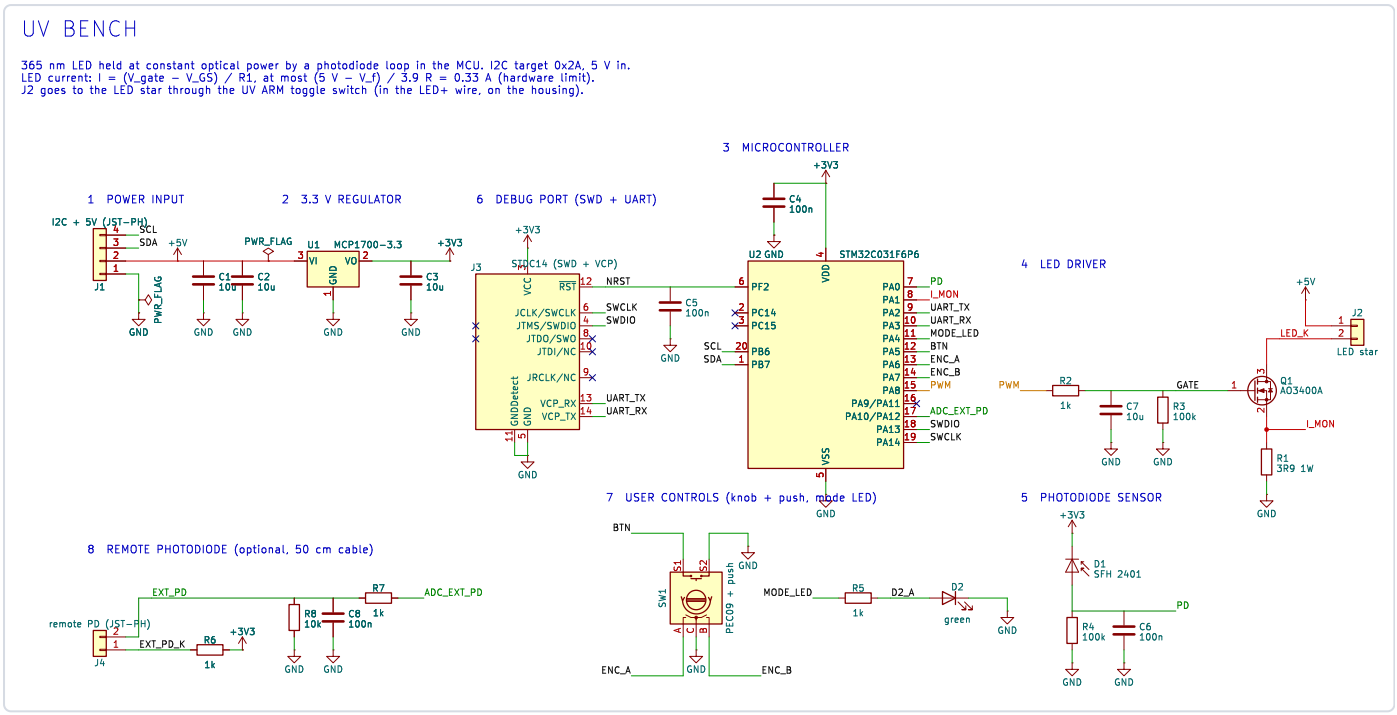


Figure 1. Rev C schematic, one sheet in 8 blocks: 1 power input, 2 3.3 V regulator, 3 MCU, 4 LED driver, 5 photodiode, 6 debug port, 7 knob and mode LED, 8 remote photodiode. The symbol reads MCP1700; the fitted part is the pin-compatible MCP1702 (stock).

6. Main components

Ref	Part	Role, and why this one
U2	STM32C031F6P6 (TSSOP-20)	Runs the loop: PWM (TIM1), 3 ADC inputs (PD, I_MON, remote PD), I2C target, encoder timer, UART. Cheap Cortex-M0+, all functions on default pins
Q1, R1	AO3400A + 3.9 Ω 1 W (2512)	Current sink. Logic-level MOSFET, fully on at 2.5 V gate; R1 sets the limit and gives a free current measurement
R2, C7, R3	1 kΩ, 10 µF, 100 kΩ	PWM → DC filter (τ = 10 ms); R3 pulls the gate low during reset, so the LED stays off
D1, R4, C6	SFH 2401, 100 kΩ, 100 nF	On-board photodiode watching the LED; V = I_{pd} × 100 kΩ, filtered at 10 ms
U1	MCP1702 3.3 V LDO (SOT-23)	3.3 V for the MCU, also the ADC reference. ~5 mA load, so a linear regulator wastes only ~9 mW
SW1, D2	Bourns PEC09 encoder + push, green LED	Manual mode. An encoder, not a potentiometer, so switching mode never makes the power jump
J1 / J4	JST-PH 4-pin / 2-pin	Host cable (GND, 5 V, SDA, SCL), keyed / remote photodiode, protected by R6, R7, C8
J3	STDC14 (Samtec FTSH-107, keyed)	Plugs straight into the ST-LINK V3MINIE: SWD programming plus a UART console over the same USB cable
J2	2 solder pads	Wires to the LED star, through the ARM switch

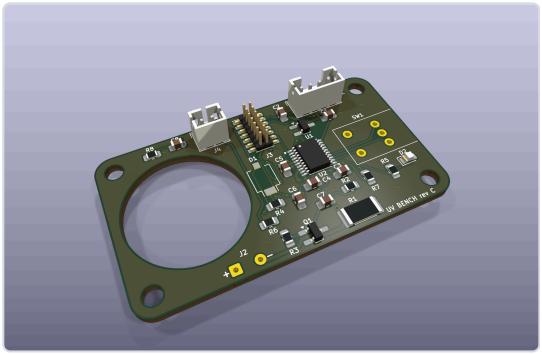


Figure 2. Main board, KiCad 3D view. The LED star sits in the round cut-out; D1 watches it from the cut-out's edge. D1 and SW1 (right) have no 3D model in KiCad's library, so only their pads show.

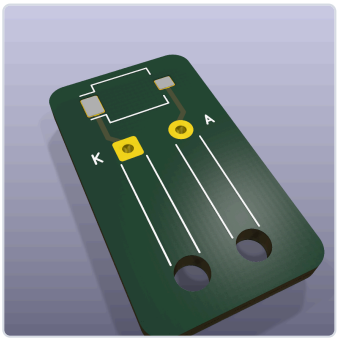


Figure 3. Sensor head, 12 × 21 mm: SFH 2401 pads, cable pads K and A with strain-relief holes.

Sources in the repository: docs/rev-b-simple.md (design notes), docs/interface-spec.md (v2.3), docs/bom-digikey.md, hardware/uv-bench and hardware/pd-head (KiCad 10). github.com/L-A-7/Electronic-design